Day-3 KNN

Given Dataset

Temperature (°C)	Fuel Consumption (L/hr)
40	12
45	14
50	15
55	18
60	20
65	22
70	25
75	27

Distance, $d = |x_i - x_8|$

Where
x_i = existing temperature
$x = 58$

As $K = 3$

Temperature (°C) x	Fuel Consumption (L/hr) y	|Distance|
55	18	2
60	20	3
65	22	7

$$\hat{y} = \frac{y_1 + y_2 + y_3}{3}$$

$$= \frac{20 + 18 + 20}{3}$$

$$= \underline{\underline{20}}$$

Predicted fuel consumption for 58°C = $\underline{\underline{20 L/hr}}$

Day3 > code_3.py > ...

```python
import pandas as pd
from sklearn.neighbors import KNeighborsRegressor
data = pd.read_csv("Day3/KNN_Dataset.csv")

x = data[["Temperature"]]
y = data["Fuel_Consumption"]

model = KNeighborsRegressor(n_neighbors=3)
model.fit(x, y)
new_data = pd.DataFrame({"Temperature": [58]})
prediction = model.predict(new_data)

print("========================================")
print("KNN REGRESSION RESULT")
print("========================================")
print("Predicted fuel consumption for a temperature of 58 degrees is:", prediction[0], "liters per hour")
print("========================================")
```

TERMINAL

```
PS C:\Users\Aleni\OneDrive\Documents\AI_ML_INTERNSHIP> & C:\Users\Aleni\AppData\Local\Programs\Python\Python312\python.exe c:/Users/Aleni/OneDrive/Documents/AI_ML_INTERNSHIP/Day3/code_3.py
Predicted fuel consumption for a temperature of 58 degrees is: 20.0 liters per hour
PS C:\Users\Aleni\OneDrive\Documents\AI_ML_INTERNSHIP> & C:\Users\Aleni\AppData\Local\Programs\Python\Python312\python.exe c:/Users/Aleni/OneDrive/Documents/AI_ML_INTERNSHIP/Day3/code_3.py
========================================
KNN REGRESSION RESULT
========================================
Predicted fuel consumption for a temperature of 58 degrees is: 20.0 liters per hour
========================================
PS C:\Users\Aleni\OneDrive\Documents\AI_ML_INTERNSHIP>
```